Status	Finished
Started	Tuesday, 20 October 2026, 11:46 AM
Completed	Tuesday, 20 October 2026, 11:49 AM
Duration	2 mins 35 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The first element is returned
d. The last element is returned

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By dividing the list length by two ✓
c. By starting from the first element
d. By comparing each element sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Sorting elements in a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **5**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 10 ✓
d. 1000

Question **6**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Sort the list ✗
- b. Divide the list into two equal parts
- c. Compare the target element with the middle element in the list
- d. Compare the target element with the first element in the list

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search stops
- c. The search continues from the beginning
- d. The search continues in the left or right sublist ✓

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. log n
- b. n ✓
- c. n-1
- d. n/2

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search stops immediately ✓
c. The search starts over from the beginning
d. The search continues until the end of the list

Question **10**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The list is sorted
c. The search ends successfully ✓
d. The search continues in the right sublist

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 99 ✓
d. 90 and 94

Question **12**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. Item is the last element in the array or item is not there at all

c. When the item is the last element in the array

d. When the item is somewhere in the middle of the array ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering

b. Searching ✓

c. Mining

d. Finding

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. There must be a mechanism to access middle element directly

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. Number values should only be present

Number values should only be present

c. The list must be sorted

d. There must be a mechanism to delete and/or insert elements in the list ✓